

Arite® BECLOMETASONE 100mcg / ACTUATION METERED DOSE INHALER

COMPOSITION:

Each actuation delivers:	
Beclometasone Dipropionate	100 mcg
Ethanol	8% w/w

This product contains a propellant, which does not contain any chlorofluorocarbons (CFCs).

PRESENTATION:

A colourless solution in a pressurized aluminium canister fitted with a metering valve and actuator, and with propellant which is CFC free.

INDICATION:

For the prophylactic management of asthma.

PHARMACOLOGY:

Mechanism of action:

Arite Beclometasone 100mcg/actuation Metered Dose Inhaler contains beclometasone dipropionate in solution in propellant HFA-134a resulting in an extrafine aerosol. The aerosol droplets are on average much smaller than the beclometasone dipropionate particles delivered by CFC suspension formulations or dry powder formulations of beclometasone dipropionate. Inhaled beclometasone dipropionate is now well established in the management of asthma. It is a synthetic glucocorticoid and exerts a topical, anti-inflammatory effect on the lungs, with fewer systemic effects than oral corticosteroids. CFC-free beclometasone dipropionate aerosol inhalers shown to exhibit equivalent therapeutic effects, pulmonary function and control of symptoms at lower total daily doses than CFC containing beclometasone dipropionate aerosol inhalers.

Pharmacokinetic:

The pharmacokinetic profile shows that the peak serum concentration for total-beclometasone (BOH) (total of any beclometasone OH and beclometasone dipropionate or monopropionate hydrolysed to beclometasone OH) after single and multiple doses is achieved after 30 minutes. The value at the peak is approximately 2ng/ml after a total daily dose of 800 mcg and the serum levels after 100, 200 and 400mcg are proportional. The principal route of elimination of beclometasone dipropionate and its several metabolites is in the faeces. Between 10% and

15% of an orally administered dose is excreted in the urine, as both conjugated and free metabolites of the drug. Lower total daily doses of CFC-free beclometasone dipropionate aerosol inhalers has shown to achieve the same clinical effect than CFC containing beclometasone dipropionate aerosol inhalers. Pharmacokinetic studies with CFC-free beclometasone dipropionate aerosol inhalers have not been carried out in any special populations.

DOSAGE AND ADMINISTRATION:

The recommended total daily dose of Arite Beclometasone 100mcg/actuation Metered Dose Inhaler is lower than that for current CFC-BDP products and should be adjusted to the individual patient. Proper instruction and good inhaler technique is necessary to get maximum benefit from inhaler. Patients should be advised that Arite Beclometasone 100mcg/actuation Metered Dose Inhaler may have a different taste and feel than a CFC inhaler. Patients should be instructed to rinse their mouth out each time after using inhaler.

Starting and Maintenance Dose:

The recommended dose of Arite Beclometasone 100mcg/actuation Metered Dose Inhaler in adults is as follows:
For mild to moderate asthma: 50mcg to 200mcg twice daily
For more severe asthma: doses up to 400mcg twice daily
Maximum recommended daily dose: 800mcg
In children aged five years and over, the recommended dose is 50mcg twice daily.

Arite Beclometasone 100mcg/actuation Metered Dose Inhaler must be used on a regular basis even when patients are asymptomatic. When patients' symptoms remain satisfactorily controlled, the dose can be gradually reduced to the minimum effective dose to maintain control. Doses of beclometasone dipropionate can be titrated up or down by switching between Arite Beclometasone 50mcg/actuation Metered Dose Inhaler (a lower strength) and Arite Beclometasone 100mcg/actuation Metered Dose Inhaler as required.

Transferring Patients from a CFC-BDP Inhaler to Arite Beclometasone 100mcg/actuation Metered Dose Inhaler:

Step 1 - Consider the dose of CFC-BDP appropriate to the patients' current condition. Symptomatic patients may require an increased dose of CFC-BDP and this increased dose should be considered in transferring patients to Arite Beclometasone 100mcg/actuation Metered Dose Inhaler.

Step 2 - Convert the appropriate CFC-BDP dose to the Arite Beclometasone 100mcg/actuation Metered Dose Inhaler dose according to the table below:

	Daily Dose of Beclometasone Dipropionate (mcg/day)						
CFC-BDP	200 - 250	300	400 - 500	600 - 750	800 - 1000	1200 - 1500	1600 - 2000
Arite Beclometasone 100mcg/actuation Metered Dose Inhaler	100	150	200	300	400	600	800

Special Patient Groups:

Elderly and Patients with Hepatic or Renal Impairment
No special dosage recommendations are made.

Patients Not Receiving Systemic Corticosteroids:

For patients who are inadequately controlled with bronchodilators and who are not receiving systemic corticosteroids, it is recommended that they continue to use a bronchodilator when treatment with Arite Beclometasone 100mcg/actuation Metered Dose Inhaler commences. Any improvement in respiratory function is usually apparent in 1 to 4 weeks. Some of the patients who do not respond during this period may have excessive mucus in their bronchi so that the drug is unable to penetrate to its site of action. A short course of systemic steroids in relatively high dosage should be given to eliminate mucus and other inflammatory changes in the lungs. Continuation of treatment with Arite Beclometasone 100mcg/actuation Metered Dose Inhaler usually maintains the improvement achieved with the oral steroid while it is being withdrawn gradually. Exacerbation of asthma caused by infection is usually controlled by appropriate antibiotic treatment and, if necessary, by increasing the dose of Arite Beclometasone 100mcg/actuation Metered Dose Inhaler. However, it may be necessary to give a short, intensive course of systemic steroids to tide over the duration of the stress.

Steroid Dependent Patients:

As recovery from impaired adrenocortical function, caused by prolonged systemic steroid therapy is slow, adrenocortical function should be monitored regularly. The patient's asthma should be in a stable state before being given inhaled steroids in addition to the usual maintenance dose of systemic steroid. Withdrawal of systemic steroids should be gradual, starting about seven days after the introduction of Arite Beclometasone 100mcg/actuation Metered Dose Inhaler therapy. For daily oral doses of prednisolone of 10mg or less, dose reduction in 1 mg steps at intervals of not less than one week is recommended. The dose reduction scheme should be chosen to correlate with the magnitude of the maintenance systemic steroid dose. Some patients feel unwell experiencing aches and pains, tiredness and even depression during the withdrawal phase despite maintenance or even improvement of respiratory function. These withdrawal symptoms should be treated symptomatically and the patient should be encouraged to persevere with the inhaler and withdrawal of systemic steroids. However, if there are objective signs of adrenal insufficiency, it may be necessary to resume systemic steroid treatment temporarily. Most patients can be successfully transferred to inhaled steroids with maintenance of good respiratory function, but special care is necessary for the first months after the transfer until the hypothalamic-pituitary-adrenal (HPA) system has sufficiently recovered to enable the patient to cope with emergencies such as trauma, surgery or severe infections. It may be advisable to provide such patients with a supply of oral steroid to use in such emergencies. The dose of inhaled steroids should be increased at this time and then gradually reduced to the maintenance level after the systemic steroid has been discontinued. Discontinuation of systemic steroids may cause exacerbation of allergic diseases such as atopic eczema and rhinitis previously controlled by the systemic drug. These should be treated symptomatically with antihistamines and / or topical therapy.

Method of administration:

This product is recommended for those patients who have demonstrated consistent good technique with co-ordinating actuation and inhalation. The patient should read the instruction leaflet before use. Before first use of the inhaler, or if the inhaler has not been used for two weeks or more, prime the inhaler by releasing two puffs into the air.

Instructions for Use:

- Take the cover off the mouthpiece.
- Breathe out as far as is comfortable and then immediately place the mouthpiece in mouth and close lips around it.
- Start to breathe in slowly and deeply through mouth and press down on the canister inside the inhaler. This releases one puff of medicine. It is important that patient carry on breathing in after the puff is released.
- Hold breath for 10 seconds, then breathe out slowly.
- If doctor has prescribed more than one puff, repeat steps (ii) to (iv) again. After use, replace the cover on the mouthpiece.

Cleaning:

For normal hygiene, the mouthpiece of inhaler should be cleaned weekly with a clean, dry tissue or cloth. Patients should also rinse their mouth with water each time after using inhaler. Do not wash or put any part of inhaler in water.

CONTRAINDICATION:

Hypersensitivity to the active substance or to any of the ingredients.

WARNING & PRECAUTIONS:

Patients should be properly instructed on the use of the inhaler to ensure that the drug reaches the target areas within the lungs. To be effective, it must be used by patients on a regular basis, even when patients do not have asthma symptoms. When symptoms are controlled, maintenance therapy should be reduced in a stepwise manner to the minimum effective dose. Inhaled steroid treatment should not be stopped abruptly. Patients with asthma are at risk of acute attacks and should have regular assessments of their asthma control including pulmonary function tests. Arite Beclometasone 100mcg/actuation Metered Dose Inhaler is not indicated for the immediate relief of asthma attacks. Patients therefore need to have relief medication (inhaled short-acting bronchodilator) available for such circumstances. Severe asthma requires regular medical assessment, including lung-function testing, as there is a risk of severe attacks and even death. Patients should be instructed to seek medical attention if short-acting relief bronchodilator treatment becomes less effective, or more inhalations than usual are required as this may indicate deterioration of asthma control. If this occurs, patients should be assessed and the need for increased anti-inflammatory therapy considered (eg. higher doses of inhaled corticosteroid or a course of oral corticosteroid). Severe asthma exacerbations should be managed in the usual way i.e. by increasing the dose of inhaled beclometasone dipropionate, giving a systemic steroid if necessary and/or an appropriate antibiotic if there is an infection, together with β -agonist therapy. Treatment with Arite Beclometasone 100mcg/actuation Metered Dose Inhaler should not be stopped abruptly. However, systemic effects of inhaled corticosteroids may occur, particularly at high doses prescribed for prolonged periods. These effects are much less likely to occur than with oral corticosteroids. Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract, glaucoma, and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children). It is important, therefore, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control of asthma is maintained. It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is slowed, therapy should be reviewed with the aim of reducing the dose of inhaled

corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist. Prolonged treatment with high doses of inhaled corticosteroids, particularly higher than the recommended doses, may result in clinically significant adrenal suppression.

Steroid Dependent Patients:
See under *Recommended Dose*

Like other corticosteroids, caution is necessary in patients with active or latent pulmonary tuberculosis. Patients should be advised to seek medical attention for review of maintenance therapy if peak flow falls, symptoms worsen or if the short-acting bronchodilator becomes less effective and increased inhalations are required. This may indicate worsening asthma.

Beclometasone dipropionate, like other inhaled steroids, is absorbed into the systemic circulation from the lungs. Beclometasone dipropionate and its metabolites may exert detectable suppression of adrenal function. Within the dose range 100-800 micrograms daily, clinical studies with beclometasone dipropionate have demonstrated mean values for adrenal function and responsiveness within the normal range. Patients should be advised that this product contains small amounts of ethanol. At the normal doses, the amounts of ethanol are negligible and do not pose a risk to patients.

Caution: The aluminium canister of this inhaler is pressurized. It should not be broken, punctured or burnt, even when apparently empty.

Use in Pregnancy And Lactation:

The potential risk of this product for humans is unknown. There is no experience of this product in pregnancy and lactation in humans, therefore the product should only be used if the expected benefits to the mother are thought to outweigh any potential risk to the foetus or neonate.

Use in Pregnancy:

There is inadequate evidence of safety in human pregnancy. Administration of corticosteroids to pregnant animals can cause abnormalities of foetal development including cleft palate and intra-uterine growth retardation. There may therefore, be a risk of such effects in the human foetus. It should be noted, however, that the foetal changes in animals occur after relatively high systemic exposure. Beclometasone dipropionate is delivered directly to the lungs by the inhaled route and so avoids the high level of exposure that occurs when corticosteroids are given by systemic routes. The use of beclometasone dipropionate in pregnancy requires that the possible benefits of the drug be weighed against the possible hazards. The drug has been in widespread use for many years without apparent ill consequence.

Use in Breast-feeding:

No specific studies examining the transfer of beclometasone dipropionate into the milk of lactating animals have been performed. It is probable that beclometasone dipropionate is excreted in milk. However, given the relatively low doses used by the inhalation route, the levels are likely to be low. In mothers breast feeding their baby the therapeutic benefits of the drug should be weighed against the potential hazards to mother and baby. There is no experience with or evidence of safety of

propellant HFA 134a in human pregnancy or lactation. However, studies on the effect of HFA 134a on reproductive function and embryo foetal development in animals have revealed no clinically relevant adverse effects.

SIDE EFFECTS:

A serious hypersensitivity reaction including oedema of the eye, face, lips and throat (angioedema) has been reported rarely. As with other inhaled therapy, paradoxical bronchospasm may occur after dosing. Immediate treatment with a short-acting bronchodilator should be initiated, this product should be discontinued immediately and an alternate prophylactic treatment introduced. Systemic effects of inhaled corticosteroids may occur, particularly with high doses prescribed for prolonged periods. These include adrenal suppression, growth retardation in children, decrease in bone mineral density and the occurrence of cataract and glaucoma. Commonly, when taking this product, hoarseness and candidiasis of the throat and mouth may occur. To reduce the risk of hoarseness and candida infection, patients are advised to rinse their mouth after using their inhaler. The adverse events are listed as below:

Infections and infestations:

- Common: Candidiasis in mouth and throat.

Immune system disorders:

- Rare: Allergic reactions, angioedema in eyes, throat, lips and face.

Endocrine disorders:

- Very rare: Adrenal suppression*, growth retardation* (in children and adolescents), bone density decreased*

Nervous system disorders:

- Uncommon: Headache, vertigo, tremor

Eye disorders:

- Very rare: Cataract*, glaucoma*

Respiratory, thoracic and mediastinal disorders:

- Common: Hoarseness, pharyngitis
- Uncommon: Cough, increased asthma symptoms
- Rare: Paradoxical bronchospasm

Gastrointestinal disorders:

- Common: Taste disturbances
- Uncommon: Nausea

Skin and subcutaneous tissue disorders:

- Uncommon: Urticaria, rash, pruritus, erythema, purpura

Musculoskeletal and connective tissue disorders:

- Very rare: Decrease bone mineral density

Psychiatric Disorders:

- Unknown: Psychomotor hyperactivity, sleep disorders, anxiety, depression, aggression, behavioural changes (predominantly in children).

*Systemic reactions are a possible response to inhaled corticosteroids, especially when a high dose is prescribed for a prolonged time.

Effects on ability to drive and use machines:
Not relevant.

DRUG INTERACTIONS:

Arite Beclometasone 100mcg/actuation Metered Dose Inhaler contains a small amount of ethanol. There is a theoretical potential for interaction in particularly sensitive patients taking disulfiram or metronidazole. Beclometasone is less dependent on CYP3A metabolism than some other corticosteroids, and in general interactions are unlikely; however the possibility of systemic effects with concomitant use of strong CYP3A inhibitors (e.g. ritonavir, cobicistat) cannot be excluded, and therefore caution and appropriate monitoring is advised with the use of such agents.

OVERDOSAGE AND TREATMENT:

Acute overdosage is unlikely to cause problems. The only harmful effect that follows inhalation of large amounts of the drug over a short time period is suppression of HPA function. Specific emergency action need not be taken. Treatment with Arite Beclometasone 100mcg/actuation Metered Dose Inhaler should be continued at the recommended dose to control the asthma; HPA function recovers in a day or two. If excessive doses of beclometasone dipropionate were taken over a prolonged period a degree of atrophy of the adrenal cortex could occur in addition to HPA suppression. In this event the patient should be treated as steroid dependent and transferred to a suitable maintenance dose of a systemic steroid such as prednisolone. Once the condition is stabilised, the patient should be returned to Arite Beclometasone 100mcg/actuation Metered Dose Inhaler by the method described.

STORAGE:

Store below 30°C. Protect from light.

**KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK**

PACK QUANTITIES:

Available in one aluminium canister fitted with a metering valve and light brown actuator fitted with white mouthpiece cover, containing 200 actuations with an actuator per box.

Further information can be obtained from doctor, pharmacist or the manufacturer.

Product Registration Holder & Manufactured By :

 **Kotra Pharma (M) Sdn. Bhd.**
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